Isac S. Lainez Sevilla

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Education

University of Puerto Rico, Mayagüez Campus Aug 2024 – Expected May 2027
M.Sc. in Physics *Mayagüez, Puerto Rico*

- **GPA:** 4.00/4.00
- **Advisors:** Dr. Henri A. Radovan and Dr. Allison Smith
- **Relevant coursework:** Introduction to Theoretical Physics I–II, Theory of Electricity and Magnetism, Introduction to Quantum Theory, General Relativity, Astrophysics, Radio Astronomy

National Autonomous University of Honduras (UNAH) Jan 2019 – Mar 2024
B.S. in Physics, Magna Cum Laude *Tegucigalpa, Honduras*

- **Final grade:** 91/100
- **Relevant coursework:** Classical Mechanics I–II, Quantum Mechanics I–II, Optics, Modern Physics, Thermodynamics and Statistical Mechanics I–II

Research Experience

M.Sc. Thesis Research Jan 2025 – Present
University of Puerto Rico, Mayagüez Campus *Advisors: Dr. H. A. Radovan, Dr. A. Smith*
Data Analysis Pipeline for Radio Pulsar Processing: Emission Studies Using Archived Observations from the Arecibo Observatory 12-m Telescope and the Green Bank Telescope. Funded by the Heising-Simons Foundation.

- Develop a reproducible data analysis pipeline for archived pulsar observations from the Arecibo 12-m Telescope and the Green Bank Telescope (GBT).
- Analyze single-pulse observations of PSR J1713+0747 to investigate pulse energy distributions, profile variability, and polarization properties.
- Apply statistical methods to characterize pulsar emission variability across multiple observing epochs.
- Document processing and analysis procedures to ensure reproducibility and support future pulsar studies.

Research Assistant, Atmospheric Humidity Variability Jun 2025 – Aug 2025
Advisor: Dr. Hector Jimenez Gonzalez *Resilience & Action Climate Research and Its Effects on Society*
Radiometer Data Analysis of Atmospheric Humidity Variability During Rainfall Events in Western Puerto Rico

- Processed radiometer data from the start of data collection through May 31, 2025, retaining observations from March 1 through September 30 of each year (spring–summer).
- Computed the water vapor mixing ratio for each observation and interpolated the data to hourly intervals (6 AM–8 PM) and 10-mb isobaric levels from 1000 to 400 mb.
- Calculated the saturation fraction for lower (850–1000 mb), lower-middle (600–850 mb), and upper-middle (400–600 mb) layers, averaged by hour of day.
- Separated rainy and dry days using Mayagüez precipitation data and produced plots comparing mean saturation fraction by hour and atmospheric layer.

Research Assistant, Large-Scale Structure of the Universe Jan 2023 – Dec 2023
Advisor: Dr. Francisco Linares *Michoacán University of San Nicolás de Hidalgo*
Application of Statistical Techniques to Study the Formation of Large-Scale Structure in the Universe

- Conducted a literature review on cosmology and statistical techniques used in cosmological analyses.
- Reproduced fundamental cosmological equations and calculations for the large-scale distribution of matter.
- Developed Python code to analyze the correlation function of galaxy convergence and tested it on

synthetic data.

Research Assistant, Hubble Tension Aug 2023 – Dec 2023
Advisor: Dr. Carlos Gabarrete *National Autonomous University of Honduras*
Investigating the Hubble Constant (H_0) Tension through Alternative Theories of General Relativity

- Reviewed textbooks, theses, and scientific articles on the H_0 tension, modified gravity, scalar-tensor theories, the ΛCDM model, and statistical data analysis.
- Analyzed and compared results from different models to assess possible explanations for, or alleviation of, the H_0 tension.

Teaching & Technical Experience

Tutor, Honduran Physics Olympiad Committee Jan 2019 – Present
Tegucigalpa, Honduras

- Designed and delivered training sessions to prepare students for national and international physics olympiads.
- Wrote the examination problems used to select the students representing Honduras.
- Mentored students in developing physics problem-solving skills.

Laboratory Instructor of Physics Aug 2024 – Dec 2025
University of Puerto Rico, Mayagüez Campus

- Developed teaching materials and laboratory procedures, and supported the professor in planning and running lab sessions.
- Guided students through experiments and evaluated their lab work, giving feedback on physics concepts.

Systems Assistant Jan 2025 – May 2025
University of Puerto Rico, Mayagüez Campus

- Worked with the PUPPI server infrastructure and observational data from the Arecibo Observatory; diagnosed, updated, and networked the servers.
- Installed and configured Docker containers with NANOGrav software for undergraduate access, and prepared server environments for future workshops.

Laboratory Instructor of Physics Jan 2022 – Aug 2023
National Autonomous University of Honduras

- Developed didactic resources and assisted the professor in laboratory, teaching, research, and outreach activities.
- Guided students and collected and analyzed laboratory data, identifying relevant statistical distributions.

Technical Skills & Languages

Programming	Python (`numpy`, `scipy`, `astropy`, `matplotlib`, `pandas`, `seaborn`), R, LaTeX
Software & tools	PSRCHIVE, DSPSR, PRESTO, TEMPO2, CASA, CARTA, Mathematica, Git/GitHub Unix/Linux, Jupyter Notebook
Languages	Spanish (native), English (upper-intermediate), German (beginner)

Conferences & Presentations

2026 *Development of a Radio Pulsar Data Analysis Pipeline for Pulsar Emission Studies with the Arecibo 12-m Telescope*
I. S. Lainez Sevilla, H. Radovan, A. Smith, Anish Roshi, Benetge Perera
Poster, NSF CARSE and NSF SpectrumX Joint Fall 2026 Center Meeting, University of Puerto Rico, Mayagüez Campus

2026 *Pulsars and Binary Systems: From the Nobel Prize to the Double Pulsar and the Future of Gravitational-Wave Astronomy*
I. S. Lainez Sevilla
Oral presentation, University of Puerto Rico, Mayagüez Campus

2026 *Emission Properties and Variability of Millisecond Pulsars: Observations of PSR J1713+0747 with the Green Bank Telescope and PSR J04374715 with the Arecibo 12 m Antenna*
I. S. Lainez Sevilla, H. Radovan, A. Smith, Anish Roshi, Benetge Perera
Oral presentation, Alpha-Cen Meeting and Salvadoran School of Astrophysics, San Salvador, El Salvador

2026 *Exploring Pulsars Through Radio Astronomy*
I. S. Lainez Sevilla
Oral presentation, National Autonomous University of Honduras, School of Physics, Tegucigalpa, Honduras

2026 *Developing a Data Analysis Pipeline for Pulsar Emission: Observations of PSR J0437−4715 and PSR B0833−45 with the Arecibo 12-m Telescope*
I. S. Lainez Sevilla, H. Radovan, A. Smith, Anish Roshi, Benetge Perera
Poster, 44th Puerto Rico Interdisciplinary Scientific Meeting, UPR Río Piedras

2026 *Single-Pulse Analysis of the Millisecond Pulsar PSR J1713+0747*
I. S. Lainez Sevilla, H. Radovan, Michael Lam
Poster, 41st Interuniversity Mathematical Sciences Research Seminar, UPR Río Piedras

2025 *Pulsars: Fundamentals, Methods of Study, and Scientific Applications*
I. S. Lainez Sevilla
Oral presentation, University of Puerto Rico, Mayagüez Campus

2024 *Statistical Techniques for Studying Large-Scale Structure in the Universe*
I. S. Lainez Sevilla, Francisco Linares
Poster, Conference on Physics, Astronomy, Climate Change, and Mathematics, Copán, Honduras

2023 *Statistical Techniques for Studying Large-Scale Structure in the Universe*
I. S. Lainez Sevilla, Francisco Linares
Oral presentation, Physics Career Week 2023, Tegucigalpa, Honduras

2023 *Investigating the Hubble Constant (H_0) Tension through Alternative Theories of General Relativity*
I. S. Lainez Sevilla
Oral presentation, School of Physics Conference, Tegucigalpa, Honduras

2020 *Problem Solving in the III Mesoamerican Physics Olympiad*
I. S. Lainez Sevilla, Carlos Gabarrete
Oral presentation, Mesoamerican Centre for Theoretical Physics (MCTP), Chiapas, Mexico

Leadership & Service

Honduran Physics Students Association (AEHFi) 2021 – Jan 2024
Co-Founder (2021 – 2024); President (Jan 2023 – Jan 2024) *Tegucigalpa, Honduras*

- Co-founded a national organization for physics students and, as president, led it and coordinated its academic, outreach, and networking activities.
- Represented Honduran physics students in national and international academic initiatives.

Vice President and Co-Founder, Honduran Optica Chapter Jan 2023 – Jan 2024
Tegucigalpa, Honduras

- Co-founded the chapter and organized talks, workshops, and networking activities for students.
- Connected students, faculty, and the broader optics and photonics community.

Delegate, International Association of Physics Students (IAPS) Jan 2022 – Jan 2024
Tegucigalpa, Honduras

- Represented the local physics community within IAPS and promoted its activities and opportunities among students.

Co-organizer, Quantum Talks in Puerto Rico Apr 2025
Mayagüez, Puerto Rico

- Developed the program and schedule; handled logistics, registration, promotion, and moderation of talks.

Co-organizer, Physics Career Week 2021 – 2023
Tegucigalpa, Honduras

- Developed the program and schedule; handled logistics, registration, promotion, and communication with invited speakers.
- Moderated talks and Q&A sessions and supported public outreach and science communication.

Member, Academic Committee, VIII Central American and Caribbean Physics Olympiad 2023
Comayagua, Honduras

- Developed examination problems and contributed to the academic design and evaluation of the competition.

Teacher Delegate, Honduran Delegation to International Physics Olympiads 2019 – 2023
Honduras

- Accompanied and supported the Honduran student delegations as teacher delegate at the Central American and Caribbean Physics Olympiad (IV–VI editions) and the Ibero-American Physics Olympiad (XXVI–XXVIII editions).
- Supported the teams throughout each competition, including the virtual editions held in 2020–2023.

Honors & Awards

Heising-Simons Foundation – Summer Research Stipend	Jun. 2026 – Jul. 2026
Fondo Dotal Scholarship	2025
Academic Excellence Award, UNAH	2020, 2021, 2023
Participant, XXIII Ibero-American Physics Olympiad	2018
First Place, Honduran Physics Olympiad	2017

Schools, Courses & Workshops

Astronomy & Physics

2026 *Summer GBO Graduate Student Residency*, Green Bank Observatory, WV
2026 *Infrared Astronomy and Radio Astronomy Workshop*, NSF Arecibo Observatory, PR
2026 *21st NRAO Synthesis Imaging Workshop*, New Mexico Tech, Socorro, NM
2025 *Hands-on Training on Reactor Operations*, University of Florida
2025 *Infrared Astronomy and Radio Interferometry*, UPR Río Piedras
2025 *ALMA Proposal*, UPR Río Piedras
2023 *Advanced Experiments in Modern Physics*, UNAH
2021 *Geometric Methods in Physics*, UNAH
2021 *Introduction to the Standard Model*, UNAH
2021 *Introduction to Cosmology*, UNAH
2021 *Introduction to General Relativity*, UNAH

Data Science & Programming

2023 *Data Science for Business, Level 2*, IBM
2023 *Databases and SQL for Data Science*, IBM
2022 *Machine Learning with Python, Level 1*, IBM
2022 *R Programming for Data Science Essentials*, IBM
2022 *Data Analysis with R Programming*, Google
2021 *Python for Data Science*, IBM
2021 *Data Visualization Using Python*, IBM
2021 *Data Analysis Using Python*, IBM
2021 *Applied Data Science with Python, Level 2*, IBM
2021 *Python Programming*, National Training Institute

Technical & Professional Tools

2023 *Introduction to Linux*; *Intermediate MS Excel*, National Training Institute
2021 *Adobe Illustrator Programming*, National Training Institute